



Presented here is a simple circuit for switching your CCTV camera outputs sequentially. It switches four video and audio channels sequentially, one at a time.

Fig. 1 shows the circuit diagram of the four-channel video and audio sequencer. It is built around NE555 timer (IC1), 4017B decade counter (IC2), four npn transistors BC547 (T1-T4), four DPDT relays (RL1-RL4) and a few other components.

IC1 is configured as an astable multivibrator. The frequency of its pulse and on/off times are determined by resistors R1 and R2, potmeter VR1 and capacitor C3. Diode D1 makes R1 the only resistor controlling

the charging of C3, while R2 and VR1 alone control its discharge through discharge pin 7. Output of IC1 from pin 3, which is a narrow, rectangular pulse train, is applied to pin 14 of IC2 (clock in), which starts counting. The second clock input (pin 13) of IC2 is grounded through switch S1. Outputs O0 through O3 of IC2 turn-on transistors T1 through T4 in a sequence, one at a time. Each transistor controls an associated relay. Each relay, in turn, switches on/off the video and audio lines coming from each camera.

Test point	Details
TP0	GND (reference for measurement)
TP1	12V
TP2	Train of pulses

Semiconductors:

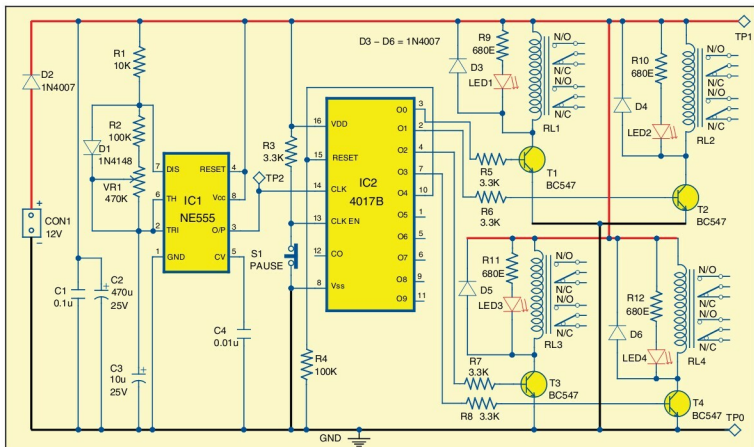
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|---|--------------------------|
| IC1 | - NE555 timer |
| IC2 | - 4017B decade counter |
| T1-T4 | - BC547 npn transistor |
| D1 | - 1N4148 switching diode |
| D2-D6 | - 1N4007 rectifier diode |
| LED1-LED4 | - 5mm LED |
| <i>Resistors (all 1/4-watt, $\pm 5\%$ carbon):</i> | |
| R1 | - 10-kilo-ohm |
| R2, R4 | - 100-kilo-ohm |
| R3, R5-R8 | - 3.3-kilo-ohm |
| R9-R12 | - 680-ohm |
| VR1 | - 470-kilo-ohm pot. |

Capacitors:

- | | |
|----|---------------------------------|
| C1 | - 0.1 μ F ceramic disk |
| C2 | - 470 μ F, 25V electrolytic |
| C3 | - 10 μ F, 25V electrolytic |
| C4 | - 0.01 μ F ceramic disk |

Miscellaneous:

- RL1-RL4 - 12V, 2C/O relay
- S1 - Push-to-off switch
- CON1 - 2-pin connector
- 12V DC supply



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